To: BirdWatcher Group

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Subject: Addressing the rise of disinformation through dynamic pre-bunking.

Executive Summary

We propose that the Birdwatcher Group develop a modular serious game that trains key societal groups in the SIFT action framework regarding lateral reading and source checking, building individual verification reflexes to counter the effects of AI-accelerated disinformation from hostile entities and overcoming the limits of reactive fact-checking.

Background

Disinformation is defined as “false information deliberately and often covertly spread to influence public opinion or obscure the truth”. While it has existed throughout history, the digital era has been uniquely intensified by generative AI and foreign influence operations from hostile countries like Russia *(EDMO, 2024)*. Empirical research indicates that “the likelihood of sharing fake news was 70% higher, due to human propensity to be attracted to novelty and emotionally arousing news” *(Vosoughit et al., 2018)*. Disinformation and misinformation can have harmful consequences on democratic society, for example, during the Covid era, approximately 800 people died and 5876 had been hospitalized due to rumors claiming that concentrated alcohol could disinfect the body *(Islam et al., 2020)*. Therefore, how can people and communities recognise and interrupt the progression from an isolated misleading claim to a wider pattern of distrust, conspiracy narratives or harmful action without granting institutions the power to dictate what citizens are allowed to believe?

Recommendations

We want to reduce the harm disinformation causes to Dutch society. We will do this by making the population aware and have the ability to recognize them. Thus, we propose to develop a serious game simulating social media and long form articles where players actively highlight unverified claims, inspect primary external records and judge the media’s credibility, to teach SIFT, lateral reading and prosocial habits.

The tool is commissioned for the Birdwatcher Group and its client network. It will be maintained and deployed through tailored modules aimed at specific target groups such as students, defense personnel, healthcare workers, and seniors. Full-scale rollout will precede high-risk civic

moments, such as political elections, ensuring resilience is established prior to real-world exposure to disinformation attacks.

Existing counter-disinformation efforts rely heavily on reactive fact-checking (debunking), however "outsized attention goes to the tangible, immediate and visible actions" *(Bateman & Jackson, 2024)*. To limit the harm of disinformation, we recommend focusing on preparing society through psychological inoculation (prebunking): training individuals to spot the disinformation and establishing a long term resilience against them *(Inoculation Science, 2025)*.

Analysis & Design choices

As previously explained, the ineffectiveness of debunking with the increasing disinformation led to the focus on prebunking instead. A study across 6 laboratory trials proves that short pre-bunking interventions significantly increase manipulative techniques detection (up to 10.5% points in real world feeds), without eroding baseline trust in verified news *(Roozenbeek et al., 2022)*, thus proving it to be a viable approach. A comparison between debunking and prebunking is shown on Table 1.1.

We decided to train the users through a gaming experience as it provides immediate reality checks to shutter the Dunning-Kruger effect, which is a bias where people with limited knowledge or competence in a given domain greatly overestimate their own competence in that domain relative to objective criteria *(Dunning, 2011)*. Another reason why we choose games is that it has been empirically proven that inoculation games work. Specifically, the studies used games to train people to detect manipulation. Results showed that the ability to recognize these techniques rose significantly, from 47% in the control group to 53%, in the game condition. Crucially, the gaming intervention increased skepticism exclusively toward manipulative claims, leaving trust in true information unaffected *(van der Linden et al., 2025)*.

However, with the increasing development and the rise of disinformation through generative AIs, training on recognizing static manipulative techniques is not enough as new techniques might raise the same issue again. To solve this component, we focused on teaching instead an actionable framework where it can be used to detect disinformation in general without focusing on the techniques.

The framework we're using is called the SIFT method: Stop, Investigate source, Find better coverage, Trace claims to original context *(Caulfield, 2019)*. This framework is based on source checking and lateral reading: "a strategy that takes the name of the website (or article, or book, etc.) you have, and searches it online to see more information in another site" *(Wineburg & McGrew, 2019)*. Similarly, studies also show that lateral reading is effective to counter disinformation, where for example in one study they showed statistically significant improvement in detecting misinformation/disinformation, with the lowest initial digital literacy individuals achieving the largest relative increase *(Nygren & Guath, 2021)*. From these researches we decided to innovatively merge them to create a game that teaches the SIFT method

and lateral reading. Finally as for the game mechanism, we focused on strongly interactive activities related to the SIFT method to keep users learning on the platform like searching for claims and deciding their validity.

Policies & Finances

Our proposal directly operationalize Pillar 1 (Maatschappelijke Weerbaarheid) of the Ministry of the Interior's (BZK) policy agenda *(Ministerie van BZK, 2022)*. It fulfills the Cabinet's mandate to build bottom-up civic digital literacy through existing public infrastructure (schools, military academies, and municipal libraries) rather than relying on state-directed content control. Additionally, it also follows Article 35 of the EU Digital Service Act that states that platforms should respect rights such as ºfreedom of expression. *(European Commission, 2022)*.

The proposed tool requires an initial cost of development, and further costs in maintenance and operation. We've given our rough estimates of costs in table 1.2, based on assumed developer effort for each scale level of our clients and their target groups. This potentially can be partially backed by government grants, provided by for example BZK *(Ministerie van BZK, 2022)* or the SIDN fund (SIDN Fonds, n.d.).

Ethics

Our educational aspect was also inspired by the ethical constraints of the problem, since education does not enforce any censorship, gives agency and promotes intellectual wellness. Furthermore, our newly developed tool simulates strictly fictional names, pictures and scenarios and is not politically charged in any way.

Challenges

The primary challenge is user participation and target group reach, failing to engage these specific demographics would result in no return from the investments. This can be mitigated by integrating the game directly into existing mandatory institutional onboarding such as schools and military academies. Another challenge is that it might train citizens to paranoically look for disinformation in the media, eroding trust in legitimate institutions. To counter this, we also added fictional verified public claims alongside deceptive ones.

Conclusion

Reactive fact-checking cannot scale against AI-accelerated disinformation. By deploying a modular serious game grounded in lateral reading and the SIFT framework, the Birdwatcher Group can proactively build resilient cognitive infrastructure across Dutch society, safeguarding democratic discourse and institutional trust without restricting freedom of expression.

Appendices

	Reactive Fact-Checking (debunk)	Automated AI fact checking (ai debunk)	Our proposal (prebunk)
Intervention timing	Arrives too late	Instant	Pre-emptive
Ethical considerations	Bias accusations	Automated censorship	Full compliance
AI disinformation adaptability	Fails	Fails	Durable

Table 1.1 Comparison between prebunking and debunking.

Scale	Reach (players)	Initial development	Annual maintenance and operation
Prototype	~200	€1k	€0.2k
Pilot	~5,000	€30k	€5k
Sector rollout	~40,000	€85k	€20k
National	~250,000	€130k	€32k

Table 1.2 Estimated costs per scale level.

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